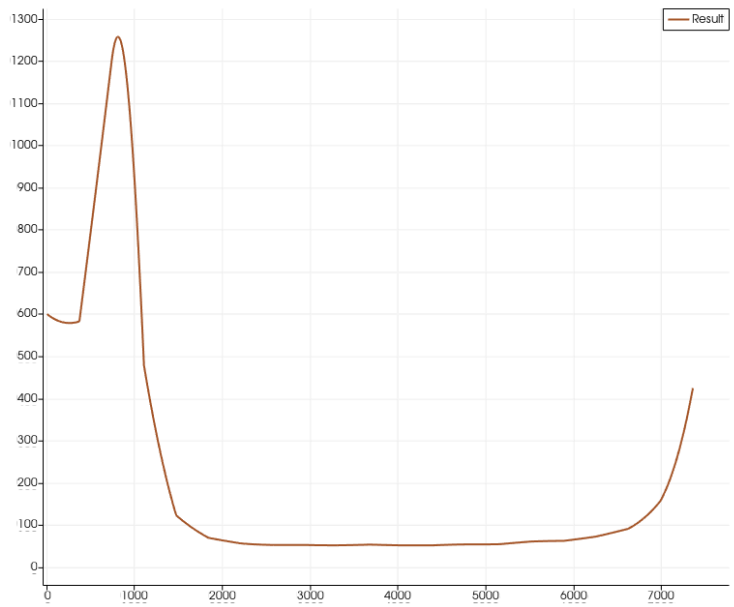
Alexandre Bismuth

Computational Medicine - Class 2B

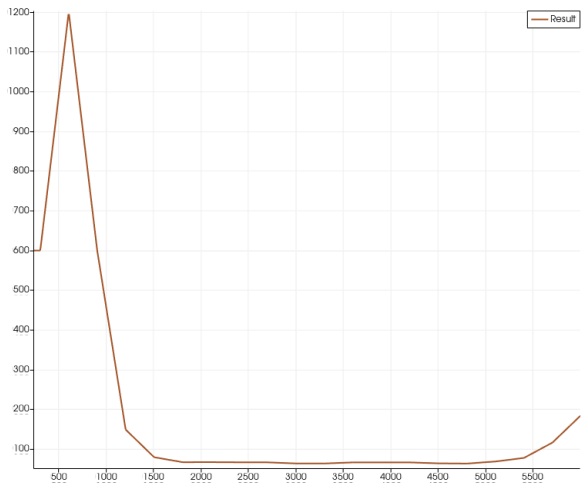
Preamble:

After plotting in a line from the top right corner to the bottom left corner following the guidance in the problem sheet, we obtain the following graph:



Task 2.1: By changing where you plot the line, measure the conduction velocity in x, y, and z direction and record it in a table as below. Does it matter where you measure – if so, why?

Here are the various plots we get depending on the direction:



X-direction plot

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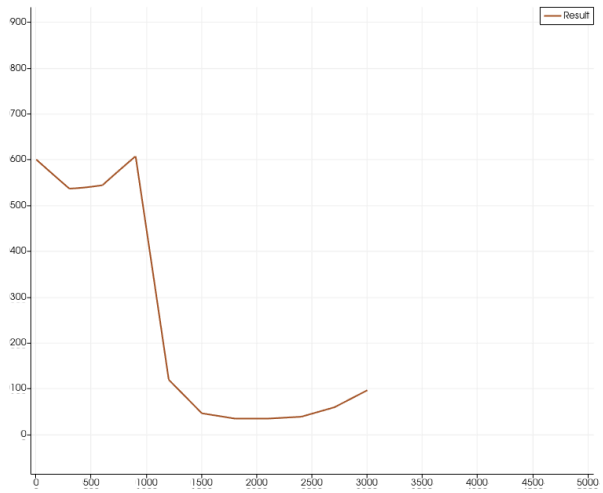
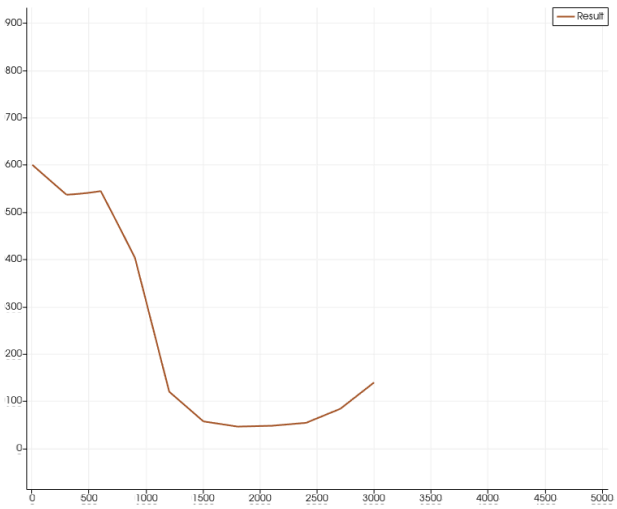


Y-direction plot



Z-direction plot

When the tissue is stimulated, it creates spikes on the sides of the graph. The true conduction velocity is in the middle of the tissue, where the wave is stable.

With 2000 elements and a resolution of 300 micrometers, we get a resolution time of 0.39 min and the following conduction velocities:

- X-velocity: 67.5 cm/s
- Y-velocity: 38.9 cm/s
- Z-velocity: 55.5 cm/s

Furthermore, we note that the XYZ direction matters because the signal along the tissue is different along the X, Y, and Z axes.

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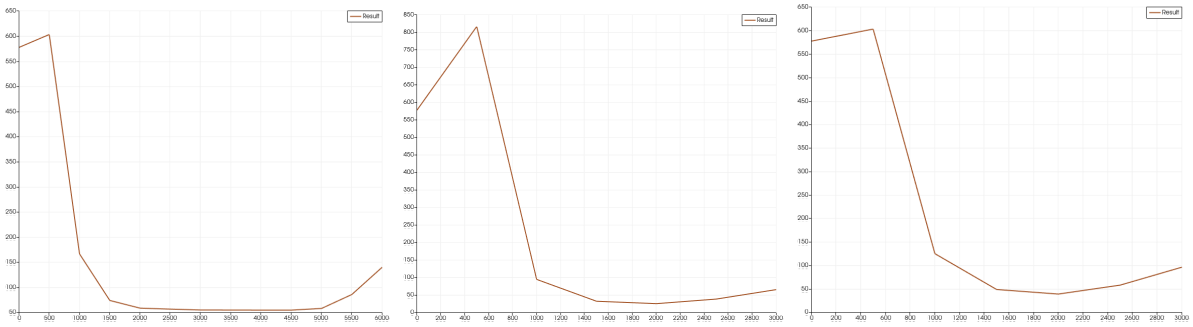
Task 2.2: Create tissue slabs with a resolution of 100 and 500 µm. Simulate in these two meshes and record the number of elements, conduction velocities, and simulation run times in the table.

For 500 micrometers, we still have 2000 elements. We get a resolution time of 0.31 minutes and the following conduction velocities in each direction:

- X-velocity: 54.3 cm/s
- Y-velocity: 32.3 cm/s
- Z-velocity: 48.9cm/s

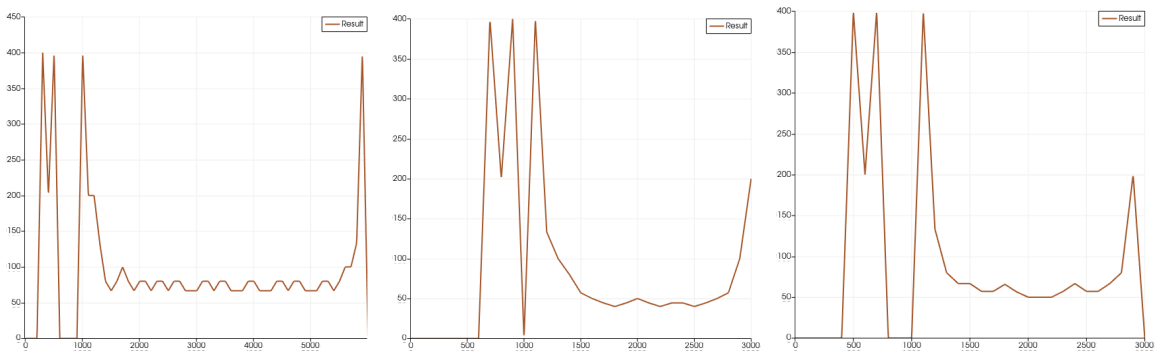
Here are the X, Y, Z (left, middle, right) plots that backup my results:



For 100 micrometers, to avoid having an incomplete slab, we choose the cube of 2000 (54 000) to keep the number of elements proportional. We get a resolution time of 3.38 minutes and the following conduction velocities in each direction:

- X-velocity: 72.5 cm/s
- Y-velocity: 47.3 cm/s
- Z-velocity: 50.0 cm/s

Here are the X, Y, Z (left, middle, right) plots that backup my results:



We can observe the abnormality of the plots for 100 micrometers. I was nonetheless unable to identify the root cause of this difference.

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What implications does changing the mesh resolution have?

Changing the mesh affects the precision with which we will render the simulation. Using a more precise resolution nonetheless requires many (with a cube relation) more elements, so computational complexity has to be heavily taken into account.

How could we achieve the same conduction velocity in different mesh resolutions?

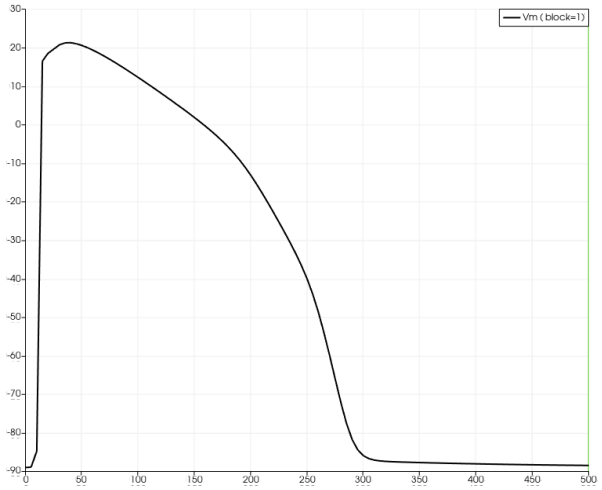
By adjusting the tissue connectivity/diffusion, we can control how fast the signal spreads and cancel out the changes in speed caused by using a coarser or finer computational grid.

Task 2.3: Which mesh resolution would you choose to simulate in and why?

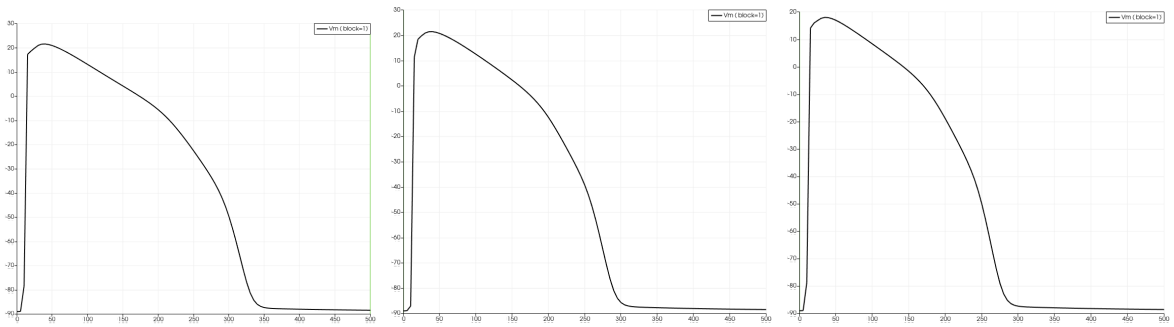
I would choose to simulate the mesh with resolution 300 as it provides a good middle ground. It is before the point where the cube number of elements that affect the resolution starts to get out of hand.

Exercise 3: Simulating ion channel block

Below is the control plot without any block



Here are plots for the modified IKr (left), INa (middle), and ICaL (right) coefficients.



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Task 3.1: Blocking which of these ion channels has the highest impact on the excitation/activation of the tissue slab? What could the physiological reason behind this be?

Blocking INa has the greatest impact on tissue excitation and activation.

This is because the sodium current is responsible for the first phase of the AP. Blocking it reduces the maximum upstroke velocity which suppresses depolarization as can be seen in the middle plot (the curve stays at about -90 for longer)

Task 3.2: Blocking which of these ion channels has the highest impact on the relaxation/repolarisation of the tissue slab? What could the physiological reason behind this be?

As we can see on the above plots, blocking the IKr coefficient has the highest impact on the repolarisation of the tissue slab. It delays repolarisation by about 50 ms.

This is because the IKr current is responsible for returning the cell to its resting potential. Reducing it disrupts the balance required for normal relaxation.

Task 3.3: In your own words, define VVUQ and discuss how it has been addressed in this exercise sheet. How could VVUQ look in other areas of computer science?

VVUQ involves ensuring the underlying mathematical model is correct, accurately represents reality, and can meaningfully capture the uncertainty associated with both inputs and outputs.

In this exercise sheet, it has been addressed in our simulations. Verification was done by comparing MonoAlg3D against MATLAB. Validation was achieved by blocking some currents to confirm the model produces biologically accurate behaviors. Finally, UQ was addressed in our study of how different resolutions affect conduction velocity.

In other areas of computer science, VVUQ can take multiple forms. In Machine Learning for example, this is exemplified by theoretical deep learning that is able to verify the correctness of the mathematical model as well as with bayesian inference which is capable of accurately capturing the model uncertainty in its outputs.